

SIDE VIEW

from the right · dashed = hidden · purple = mechanism

HEAD = WINDING KEY
turn it 28x clockwise
(seen from above)

NECK RATCHET
12 teeth + flexure
pawl: winds one way

RUBBER MOTOR
3 loops of #16 band
= 6 strands x 60 mm

BODY: 4 PLA quarter-
shells, snap-fit at
waist + side seams

2 M8 STEEL WASHERS
low on the spindle
→ CG down

TAIL = anti-tip
skid (2.5 mm up)

COUNTERSHAFT: 8 T pinion + 12 T spur
→ 12 T spur on the rear axle (A)

Ø20 REAR DRIVE WHEELS, O-ring
tyres, 1 mm eccentric hubs (C)

Ø12 front rollers
under the feet

TRIGGER PIVOT +
crank (left flipper,
far side)

LOCK ROD in a
guide (see B)

CROWN GEAR 32 T
on the motor

BASE hides the
wheels + gears

FRONT VIEW (armed)

rocks ±3.4° (C)

HEAD TURNS
on the neck
ring

TRIGGER = LEFT
flipper; raised
= ARMED (locked)

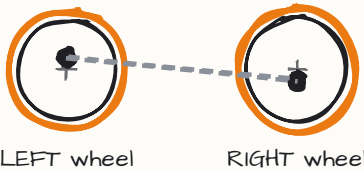
lock rod (B)

countershaft
(left side)

BASE: white
PLA chassis

48 base
93 with the flipper raised

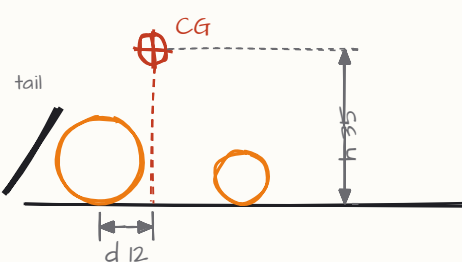
C WADDLE



axle hole 1 mm off-centre,
opposite sides (drawn x2)

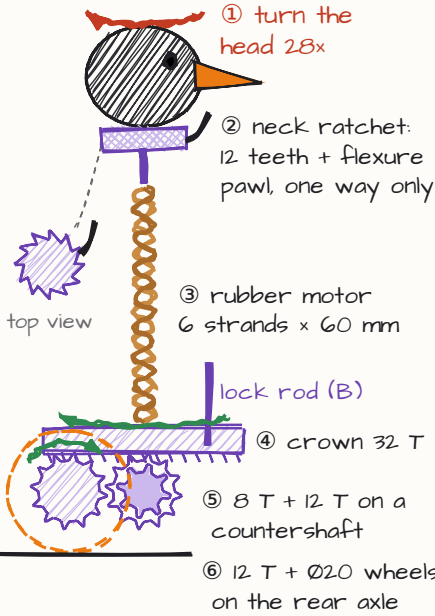
Roll = atan(2 + 34 mm track)
≈ ±3.4° every wheel turn.
One side rises as the other
drops, so the CG height stays
put: the waddle costs almost
no energy. Fast shuffle at
1 m/s, slow waddle at the end.

D WON'T TIP OVER



Launch $a = 20 \text{ mN} \div 47 \text{ g} \approx 0.4 \text{ m/s}^2$
Wheelie needs $a = g d/h$
 $= 9.81 \times 12/35 \approx 3.4 \text{ m/s}^2 \rightarrow 8\times \checkmark$
Side tip needs 21° of lean;
the waddle is only ±3.4° ✓
Backstop: the stiff tail skid lands
after 6° (emperors prop on
their tails too).

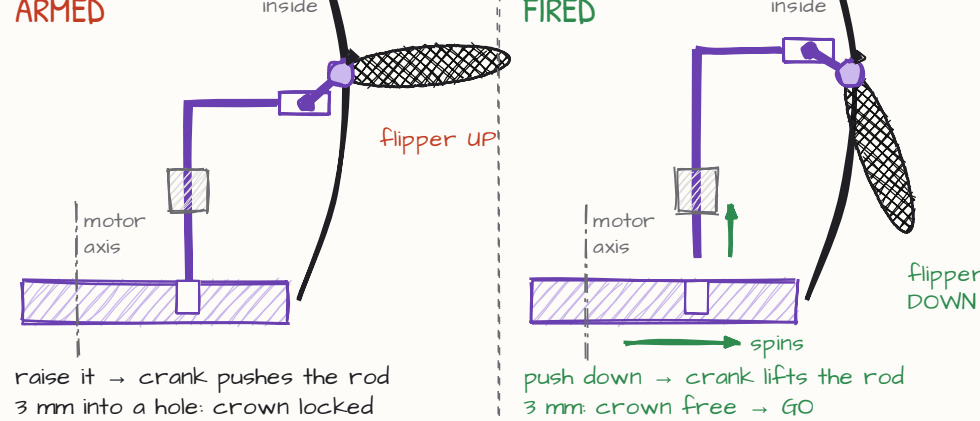
A ENERGY - TURN THE HEAD



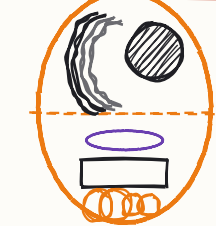
head → ratchet → rubber → crown 32 T → pinion 8 T → 12 T : 12 T → wheels (x4)

- Rubber: 3 loops of #16 band = 6 strands x 60 mm (0.51 g), hook to hook
- 28 turns (≈ 70 % of max) → $E \approx 400 \text{ J/kg} \times 0.51 \text{ g} \approx 200 \text{ mJ}$ (low-end figure for office rubber)
- Drag: rolling + 3 PLA axle bores ≈ 18 mN; printed gears $\eta \approx 0.65 \rightarrow$ needs $18 \text{ mN} \times 2.5 \text{ m} \div 0.65 \approx 71 \text{ mJ}$ (2.9x)
- 32 : 8 → 4 wheel turns per motor turn; 28 turns = 7.0 m of unwind
- Start force 38 mN (2x drag); tyre grip 0.23 N → no wheelspin
- Predicted run ≈ 6.4 m, top 11 m/s; worst case (250 J/kg) ≈ 2.8 m

B TRIGGER - FLIPPER CRANK LOCK



E PACKS INTO THE EGG



All 32 parts fit the course Kinder Maxi egg
(75 x 100 mm CAD). The body is 4 quarter-
shells that nest like bowls; Ø34 crown gear.
Assembled 109 mm tall → bigger than the egg ✓

BIOMIMICRY



Emperor penguins waddle (short legs →
side-to-side rock), stand on wide flat
feet with a low CG, and prop on stiff
tail feathers. Copied: eccentric wheels,
wide low base + steel ballast, tail
skid. Bonus: the head swivels.

HOW IT PLAYS

- 1 Raise the LEFT flipper - click:
the wheels are locked.
- 2 Turn the head 28x clockwise,
click-click... stop beak-forward.
- 3 Set it down - it stands and
waits, fully wound.
- 4 Push the flipper down → it
waddles off, ≈ 6 m.

Stop it any time: raise the flipper.

NUMBERS

- Mass ≈ 47 g, ≈ 83 % printed PLA
- CG 35 mm high, 12 mm ahead of
the rear axle (62 % on drive)
- Motor ≈ 200 mJ; needs ≈ 71 mJ
- Gears 32 : 8 + 12 : 12, module 1,
printed → x4 speed-up
- Run ≈ 6.4 m (worst case 2.8 m),
top speed ≈ 11 m/s
- Lock-rod side load ≤ 0.19 N
→ 0.06 N friction to lift
- Bought: 2 mm steel rod, 4 O-rings,
3 #16 bands, 2 M8 washers

RULES CHECK

- ✓ ≥ 75 % printed PLA (≈ 83 %)
- ✓ All parts fit the egg; the built
toy (109 mm) is bigger
- ✓ Holds its energy until the
flipper is pushed down
- ✓ Trigger is a flipper crank - no
removable pin, stays together
- ✓ No energy stored when apart
(rubber untwisted)
- ✓ Snap / press fits: no glue, no tools
- ✓ Gears are our printed parts,
not commercial toy parts

ENGG*2100 F26 · Design & Build · Concept Sketch

Sheet 2 of 2

Concept 2: EMPEROR WADDLER

Theme: Biomimicry - waddling emperor penguin
Toy type: Running toy (≥ 2.5 m)
Scale / units: mm · side view ≈ 1:1 on 11x17

Drawn by: _____ Signature: _____ Date: _____